

Model Research Document

Project: AI Workforce Displacement & Employment Impact Analysis

The dataset contains structured numerical and categorical features. The following machine learning and AI models can be implemented for prediction, classification, regression, clustering, anomaly detection and reporting.

Model	Use Case	Advantages	Expected Performance
Logistic Regression	Binary Classification	Simple, interpretable	82-87% Acc
Decision Tree	Classification	Easy to explain	84-89% Acc
Random Forest	Prediction	Reduces overfitting	92-96% Acc
XGBoost	Prediction	Excellent for tabular data	95-98% Acc
LightGBM	Prediction	Fast boosting	93-96% Acc
CatBoost	Categorical data	Handles categorical features	93-97% Acc
SVM	Classification	Works well on structured data	88-92% Acc
KNN	Similarity prediction	Simple algorithm	84-89% Acc
ANN / MLP	Complex learning	Captures nonlinear patterns	90-95% Acc
Naive Bayes	Classification	Fast baseline	80-86% Acc
Extra Trees	Prediction	High accuracy	92-96% Acc
Gradient Boosting	Prediction	Strong ensemble	91-95% Acc
AdaBoost	Classification	Boosting model	88-93% Acc
Ridge/Lasso/Elastic Net	Regression	Feature selection & regularization	High R ²
K-Means	Clustering	Customer/job segmentation	Good
Hierarchical Clustering	Grouping	Easy visualization	Good
Isolation Forest	Anomaly detection	Detects unusual cases	High Precision
PCA	Feature reduction	Reduces dimensionality	Improves efficiency
TabNet	Tabular deep learning	Interpretable DL	94-97% Acc
TabTransformer	Advanced tabular DL	Mixed feature learning	94-98% Acc
BERT	Text analysis	Skill extraction	F1:91-94%
Llama 3	Report generation	LLM insights & Q/A	Benchmark dependent

Conclusion

For this structured workforce dataset, XGBoost is recommended as the primary model because of its high predictive performance. Random Forest, LightGBM and CatBoost are excellent alternatives. Logistic Regression should be used as a baseline. K-Means and Isolation Forest can be included for clustering and anomaly detection, while BERT and Llama 3 are useful when the project includes textual analysis, report generation or an AI assistant.